

Cybersecurity & Blockchain

KGSP Enrichment [Week 3 & 4 – 2026]





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Day 03 – Securing Data

Key	
A	Back up files
B	Log off after use
C	Safe file storage
D	Passwords
E	Authorised personnel
F	Antivirus



Recap: Covered so far...

- We talked about core CIA services
 - Confidentiality
 - Integrity
 - Availability
- We talked about more security services
 - Authentication
 - Authorisation
 - Non-repudiation
- But how to achieve these services
 - Security Mechanisms

Need to secure the Data?



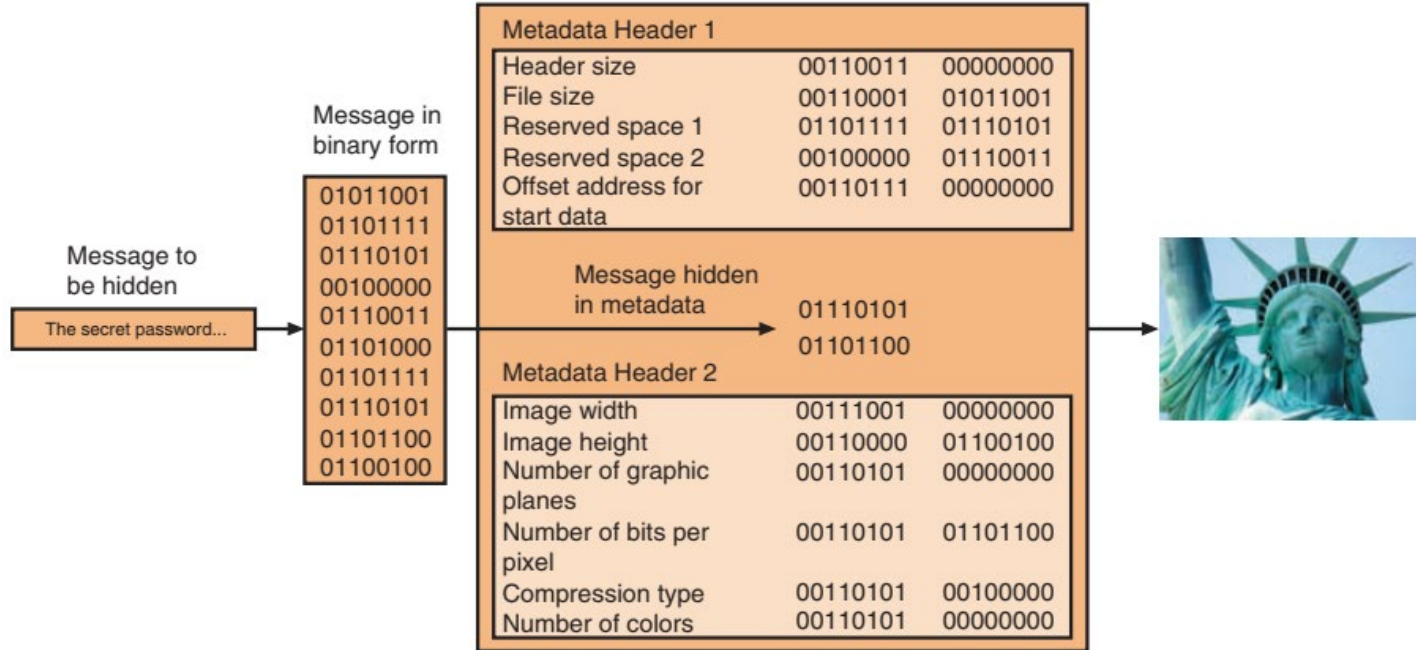
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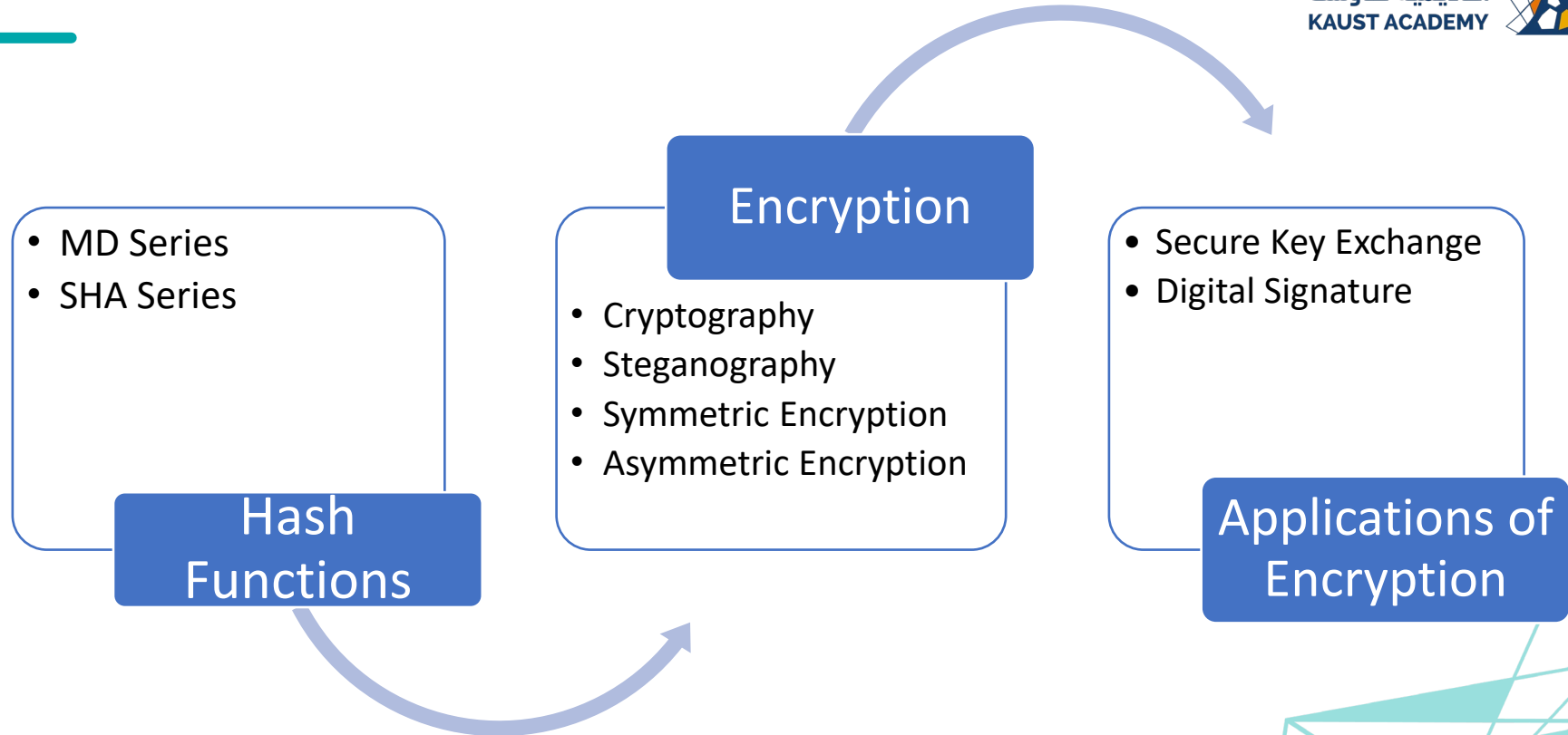
Steganography

- Steganography is the practice of **hiding secret information within a non-secret medium** in such a way that only the intended recipient knows there is hidden data
- Unlike cryptography, where the existence of the message is obvious but its content is scrambled or encrypted, steganography aims to conceal the fact that there is any hidden communication at all
- Historical account of steganography
 - Invisible Ink, Masking, Silver ball
- Modern techniques use various types of media to hide the message
 - Images (most common)
 - Audio files
 - Video files
 - Text documents

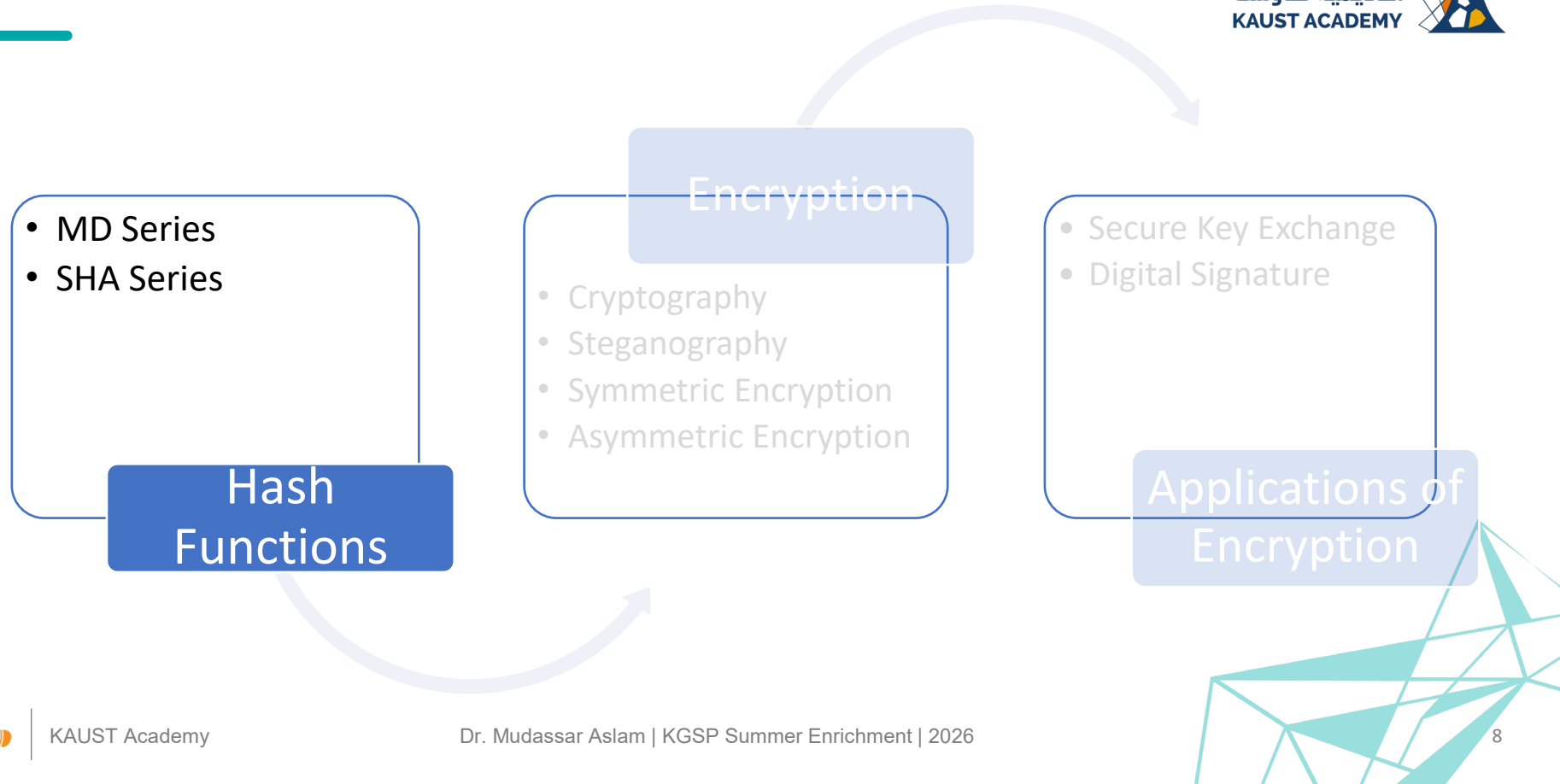
Steganography



Modern Data Security Techniques

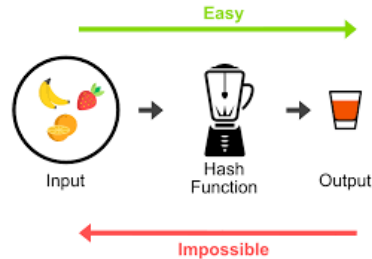


Outline

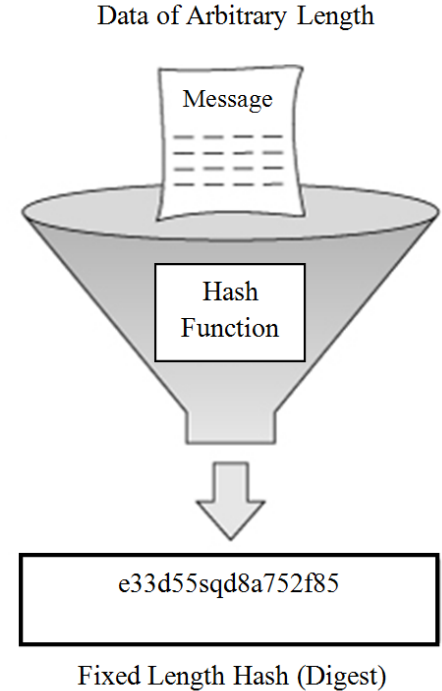
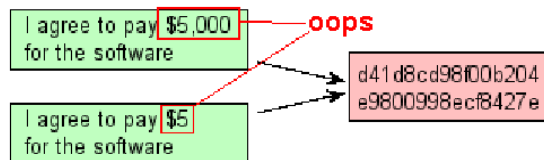


Hash function

- A one-way function that consistently transforms a variable-length input into a unique, fixed-length output.
- It can fail in one of the following two ways:
 1. If the hash function is reversible, it is not secure.



2. Someone will demonstrate that it is not collision-resistant.



Hash function – MD and SHA

▪ Message Digest 5 (MD5)

- Ron Rivest created the Message Digest 5 (MD5) hash function in 1991. That's the same Rivest who coinvented the RSA encryption algorithm.
- MD5 was the fifth in a series of hash functions
- MD5 replaced the MD4 algorithm after research showed that MD4 was insecure.
- MD5 is also considered insecure though it is still used in non-security use cases like for checksum
- MD5 produces a 128-bit hash

▪ Secure Hash Algorithm (SHA)

- Designed by NSA and published by NIST
- SHA-3 is latest and is considered secure, SHA-2 is also secure
- SHA-2 and SHA-3 can produce variable length hashes – 256 and 512 bit are common

Digests generated from one-time hash algorithms

Hash	Digest
MD2	b365b3f6ca8b35460782a658f2e82009
MD4	4fe043f7a0cde169a5d069b46bcfc0f7
MD5	7e169c6f44088e315c7b1f6513c1b0f7
RipeMD160	dd52a79bce64a1d145b51ce639e0dadda976516d
SHA-1	963ea98f0af1927e02ed0f13786162a5b8e713c0
SHA-256	f6c8a86bf6a5128cbaf2ad251b0beaa3604c11c51587de518737537800098d76
SHA-512	a2221473f8c4737d97f44265b3731a61127cc9521d4e07d18b1be357df4f04f8367a4a5255 cae956611f71b426bf494f2a68f41ca4aa122c2a07b570880e81fd
SHA3-512	3a82d58e17f3991413c5f4e9811930b69513bba02a860eed82070f892ab381f9fd926a88cf68745 565f51a93b97a1317ae8b84e2dfb798e4a2aa331187dc9e34

Hash Functions: Applications

- Check that the file we are downloading has **not been changed**
 - A software downloaded from internet
 - An important government press release
- Check the that the message or data sent over the communication network is **not changed**
 - Amount I want to pay using online banking channels has not been changed
 - Car key fob request to open the car door should be exactly as it is sent
- If someone has agreed on an online transaction, they should not be able to **deny that contract**
 - A person buys 1000 shares of a stock @\$238/share using an online trading account. The price of that stock crashes to \$205, costing him a loss of \$33000. He/she denies that he really agreed to buy the shares.

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Hash Functions: Applications

- Check that the file we are downloading has **not been changed**

- Software downloaded from internet
- An intrusion release

Storing passwords
(hash with salt)

- Check that data sent over the communication network is **not changed**

- An intrusion
- Banking channels has not been changed
- Car key
- Car door should be exactly as sent

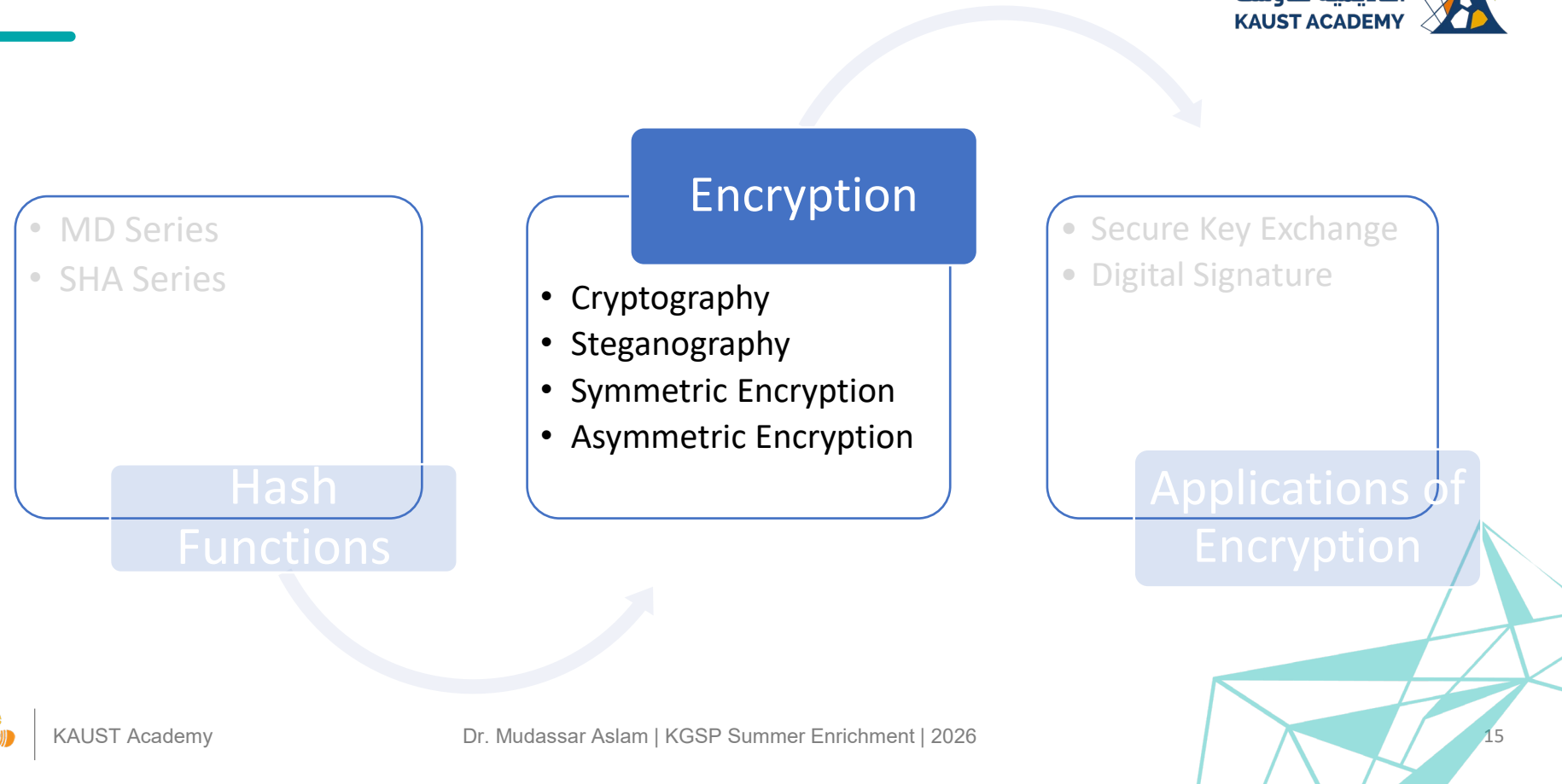
- If someone has agreed in an online transaction, they are not able to **deny that contract**

- A person buys 1000 shares of a stock & opens a trading account. The price of the stock crashes to \$205, costing him a loss of 100 shares.

Blockchain

Non-Rep

Outline



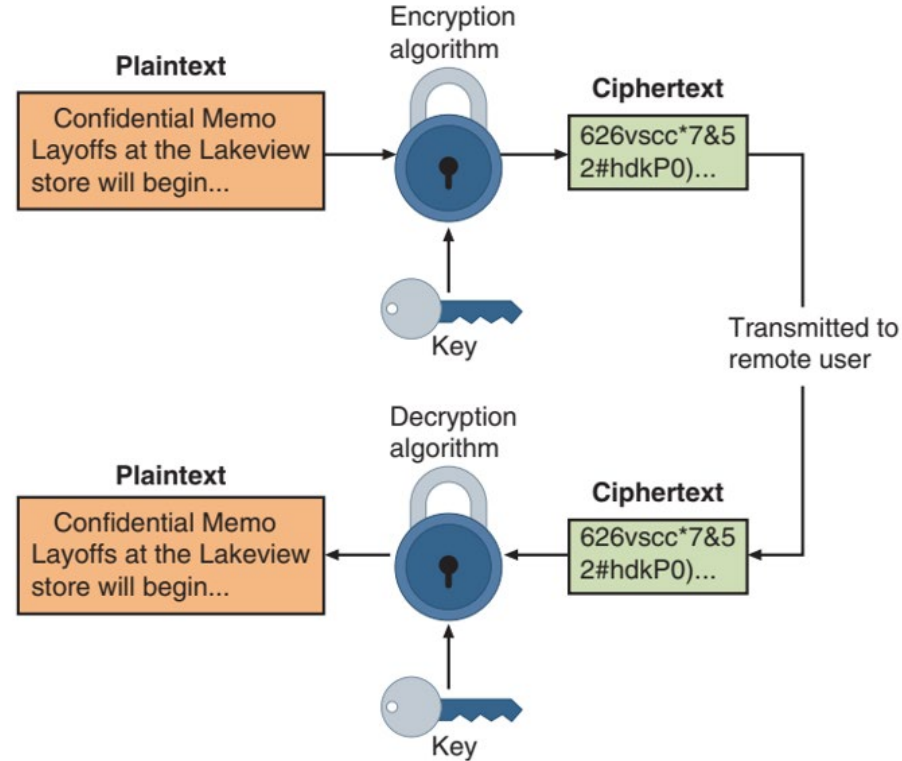
Cryptography

- The science of using encryption is called **Cryptography**
- Key concepts include
 - Encryption, Decryption
 - Symmetric encryption
 - Asymmetric encryption
 - Plain Text, Clear Text
 - Cipher Text
 - Key
 - Hashing
 - Digital Signatures

Encryption/Decryption

- Encryption protects information from prying eyes by making it unreadable to anyone who does not have the required decryption key
- **Encryption (E)** scrambles the original message/data called **Plain Text (P)** to an unreadable form called **Cipher Text (C)** by using a special technique called **Encryption Algorithm**
- Encryption algorithm uses a secret information called **Key (K)**
- **Decryption (D)** is a reverse process which brings back the original P, **if and only if**, a correct K is provided.
- Used to protect
 - Data at **rest** (e.g. stored in a computer or a storage device)
 - Data in **transit** (e.g. during communication over a network)
- **Encryption** is a security **mechanism/control** used to achieve **Confidentiality service/requirement**
- Encryption also enables **Integrity, Authenticity** and **Non-repudiation**

Encryption Process



Encryption Algorithms



SYMMETRIC
ALGORITHMS



ASYMMETRIC
ALGORITHMS



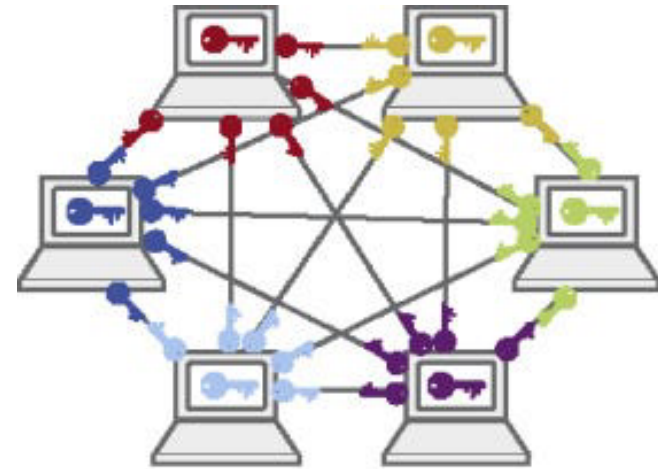
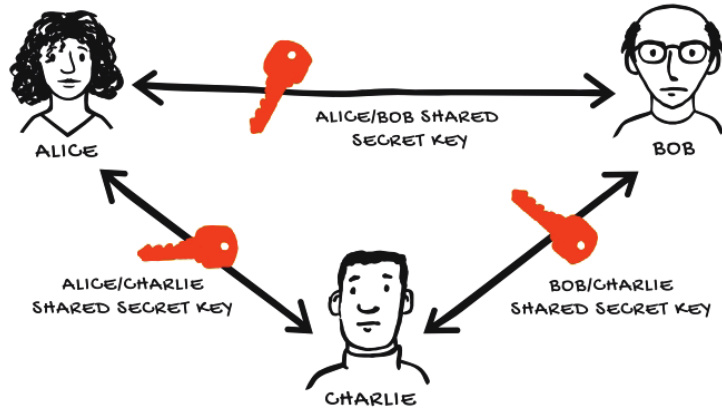
HASHING
ALGORITHMS

Symmetric Algorithms

- Symmetric encryption algorithms are also known as **shared secret** encryption algorithms
- The encryption and decryption operations use the **same key**
 - If one user encrypts a message using the secret key “**Apple**,”
 - a second user would have to decrypt the message with that same key
- It's a shared secret (e.g. a password)
- [Advanced Encryption Standard \(AES\)](#)



Symmetric encryption with more individuals



SYMMETRIC

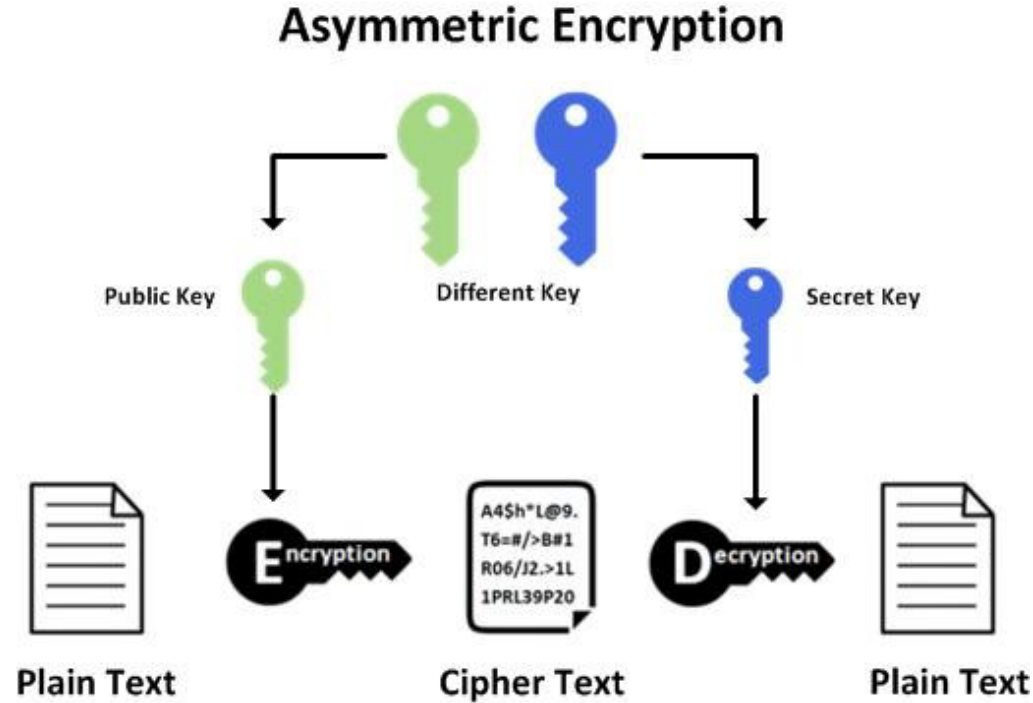
Symmetric cryptography has an equation of $\frac{n(n-1)}{2}$ for the number of keys needed. In a situation with 1000 users, that would mean **499,500 keys**.

$$3 * (3-1) / 2 = 6/2 = 3 \text{ keys}$$

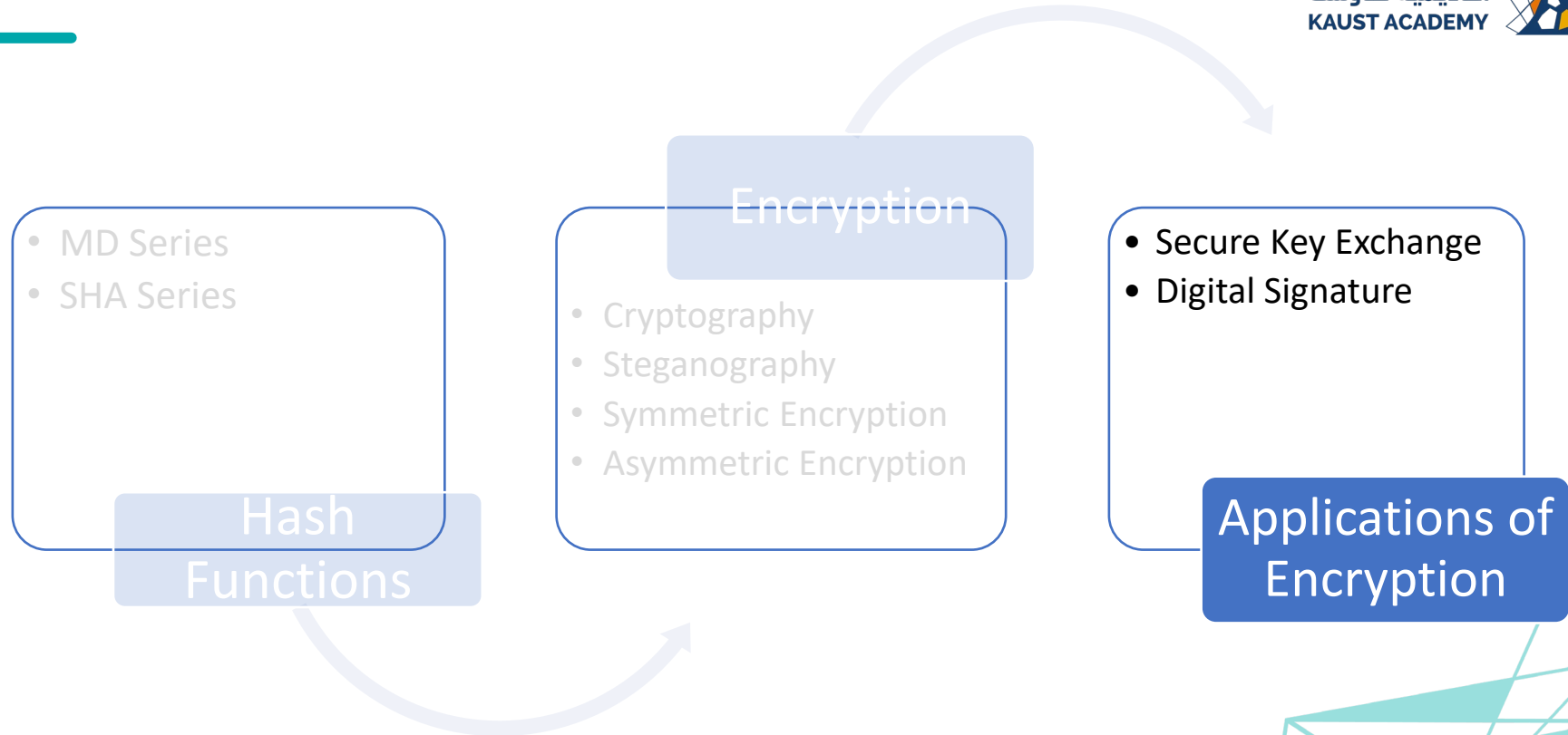
Asymmetric Encryption

- Solves the problem of scaling to large groups by using the concept of key pairs
- Each user gets two keys:
 - Public key
 - Private key
- Anything that is encrypted with one key from the pair can be decrypted with the other key from the pair.
- Slower than symmetric cryptography, but it solves the problem of creating keys for large organizations.
- Famous Asymmetric Algorithm
 - RSA
 - Diffie Hellman Algorithm
 - Elliptic Curve Cryptography (resource-constrained devices)

Asymmetric Encryption (principle)



Outline

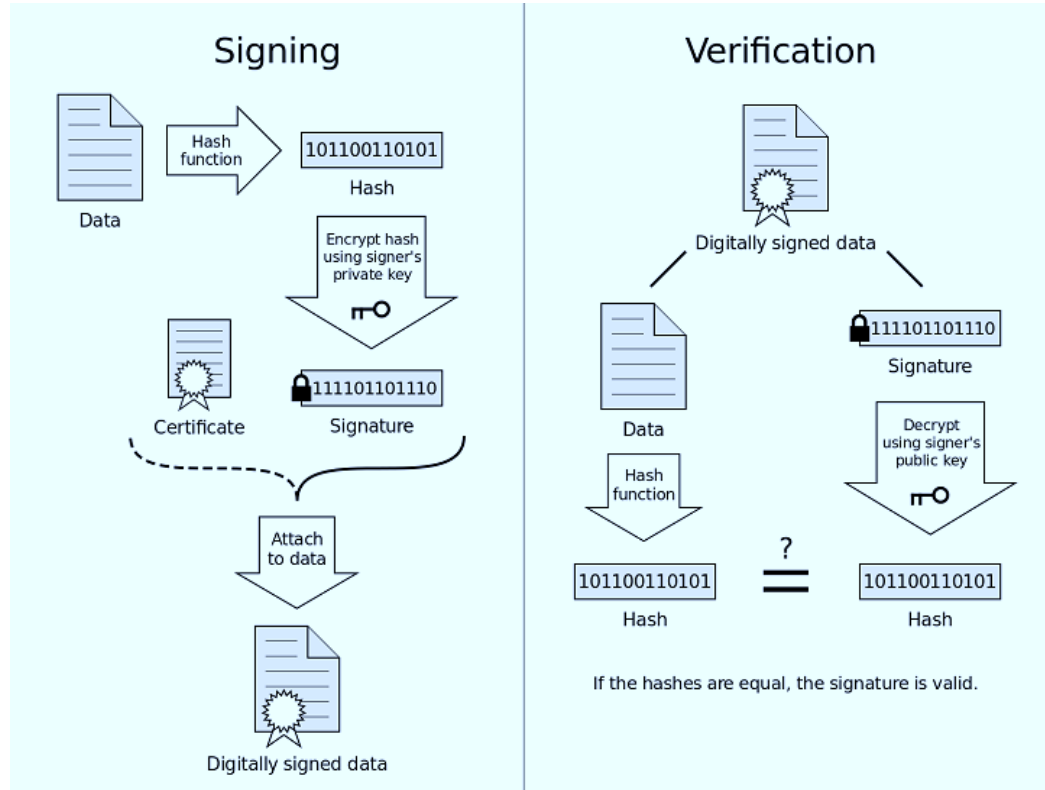


Asymmetric Encryption (Confidentiality)

Public-key encryption hides data from interceptors



Asymmetric Encryption (Digital Signature)

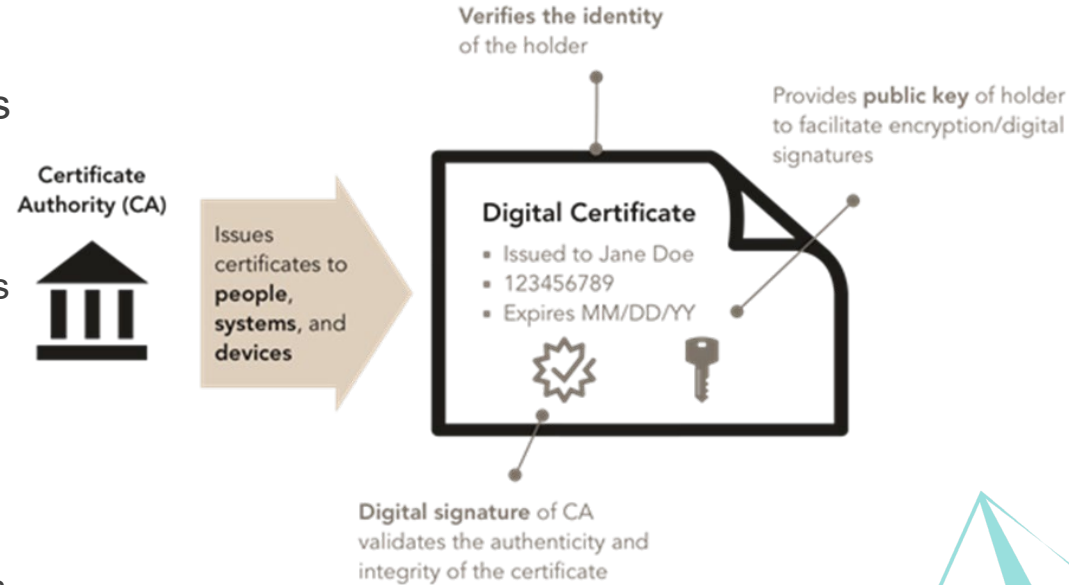


Digital Certificate – Your Internet ID

- Digital certificate is an electronic passport for the internet. It helps keep your information safe and verifies who is on the other end of your connection.

- Uses

- Proof of Identity: Verifies that a website is genuine and not a "fake" imposter site.
- The Secure Padlock: It's what creates the little padlock icon you see in your browser bar.
- Data Protection: Encrypts your private info (passwords, credit cards) so hackers can't read it.



Asymmetric cryptography practices

Action	Whose key to use	Which key to use	Explanation
Bob wants to send Alice an encrypted message.			
Alice wants to read an encrypted message sent by Bob.			
Bob wants to send a copy to himself of the encrypted message that he sent to Alice.			
Bob receives an encrypted reply message from Alice.			
Bob wants Susan to read Alice's reply message that he received.			
Bob wants to send Alice a message with a digital signature.			
Alice wants to see Bob's digital signature.			

Asymmetric cryptography practices

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Bob wants to send Alice an encrypted message.	Alice's key	Public key	When an encrypted message is to be sent, the recipient's, and not the sender's, key is used.
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Alice wants to see Bob's digital signature.	Bob's key	Public key	Because Bob's public and private keys work in both directions, Alice can use his public key to decrypt the hash.

Review: Information protections by cryptography

Characteristic	Description	Protection
Confidentiality	Ensures that only authorized parties can view the information	Encrypted information can only be viewed by those who have been provided the key.
Integrity	Ensures that the information is correct and no unauthorized person or malicious software has altered that data	Encrypted information cannot be changed except by authorized users who have the key.
Authentication	Provides proof of the genuineness of the user	Proof that the sender was legitimate and not an imposter can be obtained.
Nonrepudiation	Proves that a user performed an action	Individuals are prevented from fraudulently denying that they were involved in a transaction.
Obfuscation	Makes something obscure or unclear	By being made obscure, the original information cannot be determined.

Summary

- Encryption is the process of transforming plaintext information into ciphertext that can't be read by unauthorized individuals. Decryption uses a decryption key to convert ciphertext back into plain text.
- Encryption can be used to protect data at rest by encrypting individual files or folders. Full-disk encryption (FDE) uses encryption to protect the entire contents of a device.
- Encryption can be used to protect data in transit by encrypting it as it travels over a network. The HTTPS protocol encrypts web traffic and is a secure alternative to the unencrypted HTTP protocol. Virtual private networks (VPNs) encrypt entire network connections.
- Symmetric encryption algorithms, such as AES, use the same key to encrypt and decrypt messages. Asymmetric encryption algorithms, such as RSA, use public and private key pairs.
- Hash functions are one-way functions that transform a variable-length input into a unique, fixed-length output.



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